EIGEN · MATRIX POWERS

Four powers, computed the long way

4
POWERS

Each state is the previous one multiplied by A again. Three multiplications to reach A^4, and every one of them depends on the one before it.

$$\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \xrightarrow{\cdot A} \begin{bmatrix} 9 & 8 \\ 16 & 17 \end{bmatrix} \xrightarrow{\cdot A} \begin{bmatrix} 41 & 42 \\ 84 & 83 \end{bmatrix} \xrightarrow{\cdot A} \begin{bmatrix} 209 & 208 \\ 416 & 417 \end{bmatrix}$$

$\mathbf{A}$ $\quad A^2$ $\quad A^3$ $\quad A^4$

DETERMINANT

-5, 25, -125, 625

$\det(A^k) = (\det A)^k$

TRACE

4, 26, 124, 626

grows, and by a fixed recurrence

LARGEST ENTRY

4, 17, 84, 417

the growth the trace conceals

WHAT THE INVARIANT COLUMNS SHOW

The determinant runs $(-5)^k$ and the trace runs $5^k + (-1)^k$ — the eigenvalues of A raised to the power, and nothing else. Neither reads the entries. That is why a closed form exists at all: the growth was never a property of the sixteen numbers on display, only of the two hiding behind them.